

A counterexample to the explicit period condition in Conjecture 00000000231

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Source and scope. Both language versions define the exceptional primes by the condition that the Fibonacci period does not divide p^2 , and assert that none is below 10^{17} . Write $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_n + F_{n+1}$, and let $\pi(m)$ be the least positive period of the entire Fibonacci sequence modulo m . Under the explicit condition $\pi(p) \nmid p^2$, the prime $p = 7$ is a counterexample. The same prime works if the unspecified period is instead read as $\pi(p^2)$.

Theorem 1. *The prime $7 < 10^{17}$ satisfies*

$$\pi(7) = 16, \quad \pi(49) = 112, \quad 16 \nmid 49, \quad 112 \nmid 49.$$

Thus the source's universal nonexistence clause is false under either of the two modulus readings above.

Lemma 1 (A finite certificate for an infinite sequence). *Let $R_m(n) = (F_n, F_{n+1}) \pmod{m}$. If $R_m(k) = R_m(0)$ and $k > 0$, then $F_{n+k} \equiv F_n \pmod{m}$ for every $n \geq 0$. Conversely, any positive period k has $R_m(k) = R_m(0)$. In particular, if there is no such return at $1 \leq j < k$, then $\pi(m) = k$.*

Proof. The residue-pair recurrence is the deterministic map $(a, b) \mapsto (b, a + b \pmod{m})$. Equality of the pairs at k and 0 therefore gives equality at $n + k$ and n by induction on n . The converse uses the period equalities at $n = 0$ and $n = 1$. Any smaller positive period would consequently give a smaller return. $\square$

Proof of the theorem. Modulo 7, the residues $F_0, \dots, F_{17}$ are

$$0, 1, 1, 2, 3, 5, 1, 6, 0, 6, 6, 5, 4, 2, 6, 1, 0, 1.$$

Among the pairs with starting indices $1, \dots, 16$, the first return to $(0, 1)$ is at 16. The lemma gives $\pi(7) = 16$.

For completeness, $\pi(49) = 112$ can be checked with the same exact finite recurrence. A short certificate, for either modulus, is:

```
a, b = 0, 1
for j in range(1, k + 1):
    a, b = b, (a + b) % m
    assert ((a, b) == (0, 1)) == (j == k)
```

For $(m, k) = (7, 16)$ and $(49, 112)$ every assertion holds. The Lean project verifies every earlier pair and the terminal pair by kernel reduction of the natural-number Fibonacci recurrence, then applies the lemma. Thus this certificate establishes the least period of the entire sequence, with no assumption about unexamined indices. The supplied Python script independently recomputes both residue orbits as a supplementary check.

Finally 7 is prime, $7 < 10^{17}$, and neither 16 nor 112 divides 49, proving both stated counterexamples. $\square$

Relation to the conventional problem. The source’s divisibility condition is different from the conventional exceptional-prime condition $\pi(p^2) = \pi(p)$; see Elsenhans and Jahnel [1], Section 2.2. Indeed, the calculations above give $112 \neq 16$, so 7 is *not* an exceptional prime in that conventional sense. This submission only disproves the explicit bilingual statement. It makes no claim about the existence or infinitude of conventional Wall–Sun–Sun primes or the validity of searches for them. The later discriminant and infinitude clauses are unnecessary to the disproof of the first conjunct.

Formalization and reproduction. The Lean 4.19.0 project uses only its bundled standard library. It defines the actual Fibonacci sequence on all natural indices, proves its initial values, recurrence and uniqueness, and defines a period by a universal equality over every index. The generic least-period certificate proves the bridge from finite residue pairs to this unbounded definition. Actual natural-number primality, divisibility, leastness and the bound 10^{17} appear in the final counterexample theorems. Both modulus readings and the inequality of the conventional periods are formally proved. Kernel-checked `decide` certifies only the finite arithmetic; no oracle computation is used. No `sorry`, `admit`, `native_decide` or added axiom occurs. From `lean/`, run `lake build` and `lake env lean -DwarningAsError=true Main.lean`. Build logs, printed axiom dependencies and a solo adversarial review accompany this submission. No independent-review claim is made.

References

- [1] A.-S. Elsenhans and J. Jahnel, *The Fibonacci sequence modulo p^2 —An investigation by computer for $p < 10^{14}$* , 2010. arXiv:1006.0824.